Spherical Transformer Dynamics

Equilibria and Recurrent Dynamics

Normalized attention, serial quadratic MLPs, and continuous-depth limits

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Volume I — Attention without an MLP. Necessary and sufficient condition for all equilibria, finite layer and continuous time, spectral–Gram reduction in the case $Q^\mathsf{T}K = V = V^\mathsf{T}$, polygonal forms, precise comparison with Altafini, and a stable internal cycle produced by attention alone when score and value are untied.

Volume II — Serial attention-then-MLP block. Exact equations for all fixed points and cycles, taxonomy of four families, audit down to layers 4,096 times weaker, continuous cycles, chaos and hyperchaos, multihead maps, many-token limit, alternative kernels, observables, MLP training, a first Muon step, and slow OU weights.

Exact identities, asymptotic statements, and numerical observations are identified separately. The finite census contains 214,528 independently sampled models and 2,844,160 initial trajectories. Recurrent continuous dynamics is assessed by direct integration, tangent spectra, perturbations, return-time scaling, and basin tests.

Anonymous submission — sources, scripts, and machine-readable aggregates are included in the working dossier.